

HW Ex 1, 2, 8, 15, 20, 31

- Subspaces
- Column & Null spaces
- Basis
- Find a basis

§2.8 Subspaces of $\mathbb{R}^n$

Important subsets of $\mathbb{R}^n$: Subspaces

- Examples:
- solution sets of homogeneous linear systems
 - range of a lin. map
 - span of vectors

Def. A subspace of $\mathbb{R}^n$ is a subset H s.t.

a. $\vec{0} \in H$

b. $\forall \vec{u}, \vec{v} \in H, \vec{u} + \vec{v} \in H$ closed under +

c. $\forall \vec{u} \in H, c \in \mathbb{R}, c\vec{u} \in H$ scalar mult

Exmp 1. $\vec{v}_1, \vec{v}_2 \in \mathbb{R}^n$, $H = \text{Span}\{\vec{v}_1, \vec{v}_2\}$, then H is a subspace.

• $\vec{0} = 0\vec{v}_1 + 0\vec{v}_2 \Rightarrow \vec{0} \in H$

• $\forall \vec{u}, \vec{v} \in H$, write $\vec{u} = s_1\vec{v}_1 + s_2\vec{v}_2$, $\vec{v} = t_1\vec{v}_1 + t_2\vec{v}_2$

Then $\vec{u} + \vec{v} = (s_1\vec{v}_1 + s_2\vec{v}_2) + (t_1\vec{v}_1 + t_2\vec{v}_2) = (s_1 + t_1)\vec{v}_1 +$

Thus, $\vec{u} + \vec{v} \in H$. (s_2 + t_2)\vec{v}_2

• $\forall \vec{u} \in H, c \in \mathbb{R}, c\vec{u} = c(s_1\vec{v}_1 + s_2\vec{v}_2)$

Then $c\vec{u} = cs_1\vec{v}_1 + cs_2\vec{v}_2 \in H$

Exmp 2 $\vec{v}_1, \dots, \vec{v}_p \in \mathbb{R}^n$

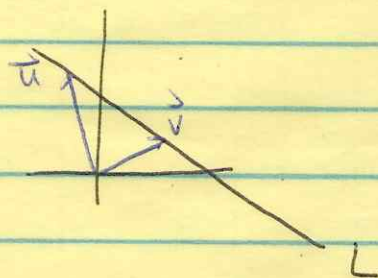
$\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\} \rightarrow$ a subspace. called the subspace

spanned by $\vec{v}_1, \dots, \vec{v}_p$

(or generated)

Exmp 3 (Nonexample) A line L in $\mathbb{R}^n$ not through the $\vec{0}$.

$\rightarrow$ NOT a subspace.



• $\vec{0} \notin L$

• $\vec{u} + \vec{v} \notin L$

• $c\vec{u} \notin L$

Column space and null space of a matrix A $m \times n$ matrix

Defn The column space of A is the set $\text{Col } A$ of all the lin. comb. of the cols of A .

In other words, $A = [\vec{a}_1 \dots \vec{a}_n]$ $m \times n$

then $\text{Col } A = \text{Span} \{\vec{a}_1, \dots, \vec{a}_n\}$. Subspace of $\mathbb{R}^m$

Exmp 4 $A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}$, $\vec{b} = \begin{bmatrix} 3 \\ 3 \\ -6 \end{bmatrix}$

Q $\vec{b} \in \text{Col } A$?

Sol Solve the equation $A\vec{x} = \vec{b}$.

If consistent then $\vec{b} \in \text{Col } A$

If not then $\vec{b} \notin \text{Col } A$

[a]

Defn The null space of A is the set $\text{Nul } A$ of all solutions of $A\vec{x} = \vec{0}$. Subspace of $\mathbb{R}^n$

Thm 12 $\text{Nul } A \subset \mathbb{R}^n$ is a subspace. (Solution set of homogeneous equation)

If check $\vec{0} \in \text{Nul } A$, closed under $+$ & scalar mult.

Left as an ex.

[a]

★ Basis for a subspace

We want to express all vectors in a subspace using the least number of vectors

Defn A basis for a subspace H of $\mathbb{R}^n$ is a linearly independent set in H that spans H .

Exmp 5 Standard basis of $\mathbb{R}^n$: $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$, ...,

$$\vec{e}_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

- linearly independent
- Span $\mathbb{R}^n$

Exmp. 6 (Null space) Find a basis for $\text{Nul } A$.

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Sol Write the sol. of $A\vec{x} = \vec{0}$ in parametric vector form:

$$[A \ 0] \sim \begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{free variables } x_2, x_4, x_5$$

General solution

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 3 \\ 0 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

all the vectors in $\text{Nul } A$

Note that $\vec{u}, \vec{v}, \vec{w}$ are linearly independent and they span $\text{Nul } A$.

Thus $\{\vec{u}, \vec{v}, \vec{w}\}$ is a basis of $\text{Nul } A$.

(REF) Exmp. 7 (Column space) Find a basis for $\text{Col } B$.

$$B = \begin{bmatrix} 1 & 0 & -3 & 5 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

p.c. p.c. p.c.

Sol Notice that all columns are linear comb of pivot cols. p.c. are linearly independent.

Thus they form a basis.

Exmp 8 (Col. space, general matrix) A : matrix

Linear dependence relations among columns of A can be expressed in the form $A\vec{x} = \vec{0}$.

Row operations does not change the solutions.

Writing a vector as a linear comb. of other vectors is a linear dependence relation

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 & -9 \\ -2 & -2 & 2 & -8 & 2 \\ 2 & 3 & 0 & 7 & 1 \\ 3 & 4 & -1 & 11 & -8 \end{bmatrix} \sim B \text{ as in Exmp 7.}$$

$\vec{a}_1 \quad \vec{a}_2 \quad \vec{a}_3 \quad \vec{a}_4 \quad \vec{a}_5$

$\uparrow \quad \uparrow \quad \quad \quad \uparrow$

p.c. p.c. p.c.

Thus, $\{\vec{a}_1, \vec{a}_2, \vec{a}_5\}$ forms a basis of $\text{Col } A$.

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Generally,

Thm 13 The pivot columns of a matrix A form a basis of $\text{Col } A$.

Warning Use the pivot columns of A , NOT its REF.